

Assessing and Improving Punctuation Robustness in English-Marathi Machine Translation

Kaustubh Shivshankar Shejole, Sourabh Deoghare and Pushpak Bhattacharyya



Computation for Indian Language Technology (CFILT)
Indian Institute of Technology Bombay, Mumbai, India



Introduction

- Neural Machine Translation (NMT) systems rely heavily on explicit punctuation cues to resolve semantic ambiguities in a source sentence.
- Inputting user-generated sentences, which are likely to contain missing or incorrect punctuation, results in fluent but semantically disastrous translations.
- This work attempts to highlight and address the problem of punctuation robustness of NMT systems through an English-to-Marathi translation.
- First, we introduce **Viram**, a human-curated diagnostic benchmark of 54 punctuation-ambiguous English-Marathi sentence pairs to stress-test existing NMT systems.
- Second, we evaluate two simple remediation strategies: cascade-based *restore-then-translate* and *direct fine-tuning*. Our experimental results and analysis demonstrate that both strategies yield substantial NMT performance improvements.
- We find that current Large Language Models (LLMs) exhibit relatively poorer robustness in translating such sentences than these task-specific strategies, thus necessitating further research in this area.

Problem Statement

To assess and improve punctuation robustness in English-Marathi machine translation.

The Viram Benchmark

- Kirkman (2006) [1] analyze punctuation in the English language, examining how ambiguity can arise from the omission of punctuation marks. For instance, in the sentence, ‘As the machine develops the forms we use to record data from past projects will be amended,’ readers must insert a comma after *develops* to derive the intended meaning. This example illustrates the human ability to extract meaning from syntactically ambiguous sentences.
- We manually curated English sentences from this work and, with the assistance of two native Marathi speakers, translated the English (Meant) sentences into Marathi.
- The resulting diagnostic benchmark comprises 54 punctuation-ambiguous instances, structured as English (Written) – English (Meant) – Marathi (Meant).
- While the benchmark size is relatively modest, it is commensurate with the significant challenges inherent in data acquisition and curation within this specific domain. Despite this, the rigor applied to its curation ensures that it serves as a high-quality, representative sample for diagnostic evaluation.

Motivation

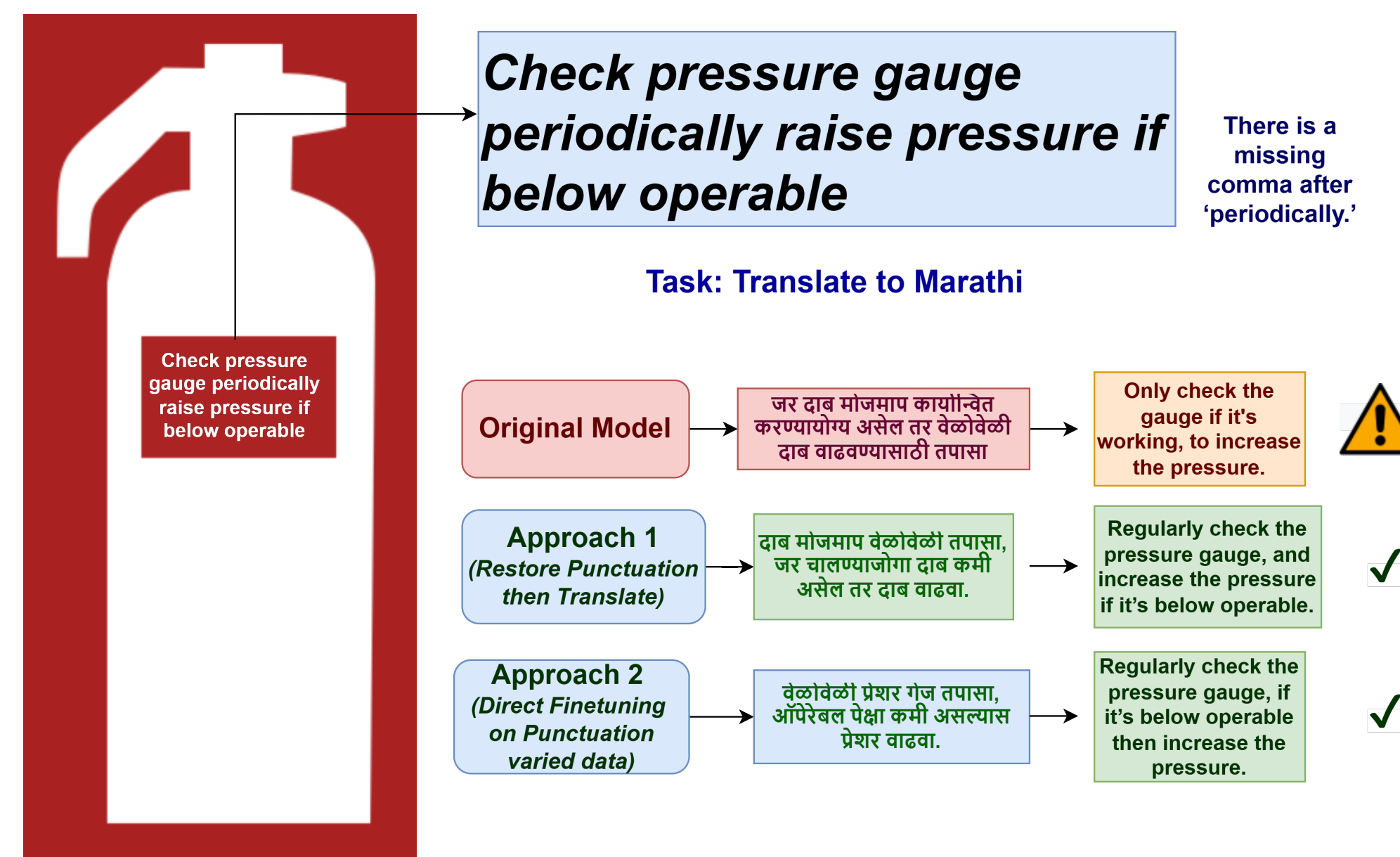


Figure 1: A missing comma can cause semantic disaster.

Methodology

Approach 1: Restore Punctuation then Translate

Punctuation is first restored in the English source text before translation. We explore two modeling paradigms for punctuation restoration:

- Token Classification:** `bert-large-uncased` and `microsoft-mpnet-base` are used to treat punctuation prediction as a sequence labeling task.
- Text-to-Text Generation:** We fine-tune `google-t5-base` to generate punctuated text from unpunctuated input and also evaluate AI4Bharat’s Cadence model without fine-tuning.

Approach 2: Direct Translation

We improve MT robustness by fine-tuning IndicTrans2 (200M) on four variants of an internal professionally curated dataset:

- With Punct:** Original data with punctuation.
- Without Punct:** Source punctuation removed.
- Combined 2x:** Original + punctuation-removed data.
- Combined x:** Alternating sentences with and without punctuation.

(“x”: the size of the internal fine-tuning dataset.)

Prompting sub-10B LLMs

We evaluate punctuation-ambiguous English-Marathi translation using zero-shot and three-shot prompting with three LLMs: **Sarvam-2b-v0.5** (2B), **Gemma-2-9b** (9B), and **LLaMA-3.1-8B** (8B). All models were exposed to Indian languages during pretraining, including Marathi. We evaluate two prompting strategies: **restore punctuation then translate** and **direct translation**. For correctly punctuated inputs, we use direct translation. For three-shot prompting, three Viram examples (comma, semicolon, colon ambiguity) are used and excluded from evaluation to ensure fair assessment.

Quantitative Analysis

Type	Model Name	BLEU	BLEURT-20	COMET	chrF++	chrF2++	LabSE	MuRIL
Baseline	IndicTrans2 on indic 200M (Original Model)	21.72	0.7916	0.7391	59.45	55.38	0.9126	0.7619
Upper Performance Boundary	IndicTrans2 on indic 200M + Input as ‘sentence meant’	26.20	0.8082	0.7606	61.15	57.41	0.9313	0.7915
Approach 1	Fine-tuned bert-large-uncased + Original Model	23.84	0.7955	0.7595	60.02	56.11	0.9199	0.7806
	Fine-tuned microsoft-mpnet + Original Model	25.12	0.7996	0.7597	60.56	56.79	0.9210	0.7813
	Fine-tuned t5-base + Original Model	24.74	0.7977	0.7586	60.68	56.92	0.9230	0.7838
	AI4Bharat’s cadence + Original Model	23.44	0.7980	0.7516	60.49	56.69	0.9210	0.7809
Approach 2	Fine-tuned (w/ punct) (x)	21.21	0.7830	0.7426	58.90	54.72	0.9145	0.7685
	Fine-tuned (w/o punct) (x)	24.66	0.7774	0.7417	60.30	56.56	0.9122	0.7830
	Fine-tuned (with and w/o punct) (2x)	24.27	0.7785	0.7443	60.61	56.83	0.9120	0.7794
	Fine-tuned (alternate with and w/o punct) (x)	24.28	0.7745	0.7433	60.21	56.52	0.9047	0.7761
LLM	GPT-5-mini (Zero-Shot + Direct Translation)	18.09	0.7786	0.7420	52.50	48.82	0.9006	0.7394
	DeepSeek-V3.2 (Zero-Shot + Direct Translation)	23.41	0.7858	0.7590	58.48	54.82	0.9197	0.7765

Oracle sentence boundaries provide the highest performance, confirming that accurate punctuation is critical for translation quality. Pipeline-based restoration (Approach 1), especially with **mpnet** and **t5-base**, closely approaches oracle performance and consistently outperforms direct fine-tuning. Fine-tuning improves robustness, while LLMs, though competitive in semantic similarity, remain below specialized pipeline systems.

Analysis on Prompting LLMs

Prompting Strategy	Approach	Model Name	BLEU	BLEURT-20	COMET	chrF++	chrF2++	LabSE	MuRIL
Zero-Shot	Restore then Translate	Llama 3.1 8b	5.16	0.6548	0.6026	37.13	33.18	0.8030	0.6093
		Gemma 2 9b	11.69	0.7260	0.6783	45.89	41.98	0.8783	0.6952
		Sarvam 2b v0.5	9.97	0.6961	0.6736	42.93	38.86	0.8228	0.6468
	Direct Translation	Llama 3.1 8b	6.37	0.6660	0.6192	39.89	35.76	0.8230	0.6314
		Gemma 2 9b	8.94	0.7248	0.6889	45.40	41.33	0.8559	0.6712
		Sarvam 2b v0.5	5.67	0.6300	0.6392	37.47	33.67	0.8102	0.6149
3-Shot Prompting	Restore then Translate	Llama 3.1 8b	4.56	0.6641	0.6113	37.80	33.68	0.8194	0.6120
		Gemma 2 9b	14.84	0.7407	0.6889	49.27	45.50	0.8926	0.7056
		Sarvam 2b v0.5	8.55	0.7352	0.6982	44.48	40.02	0.8509	0.6710
	Direct Translation	Llama 3.1 8b	7.98	0.6776	0.6272	43.21	38.91	0.8216	0.6192
		Gemma 2 9b	10.99	0.7388	0.6938	48.98	44.88	0.8883	0.7013
		Sarvam 2b v0.5	11.01	0.7571	0.7224	49.27	44.74	0.8803	0.7019
Direct Translation using Sentence Meant (Original)	Direct Translation	Llama 3.1 8b	7.87	0.6785	0.6330	39.70	35.84	0.8404	0.6535
		Gemma 2 9b	13.75	0.7256	0.6941	45.85	42.47	0.8829	0.6896
		Sarvam 2b v0.5	10.36	0.6866	0.6725	41.30	37.63	0.8422	0.6550

LLM performance improves with three-shot prompting, with **Gemma-2-9b** and **Sarvam-2b** achieving the strongest results among evaluated models. Despite improvements, all LLM configurations remain substantially below task-specific MT systems and oracle settings.

Conclusion and Future Work

- Introduced **Viram**, a diagnostic benchmark for punctuation-ambiguous English-Marathi MT. Highlighted the problem of punctuation sensitivity in English-Marathi NMT and evaluated two simple strategies: *restore-then-translate* and *direct fine-tuning*, both improving robustness over the baseline. Sub-10B LLM prompting performance remains below task-specific systems.
- Future work: extending to other Indic languages and design evaluation metrics that better capture meaning preservation and exploring architectures that handle punctuation ambiguity natively, reducing reliance on multi-stage pipelines.

References

- [1] John Kirkman. *Punctuation matters: Advice on punctuation for scientific and technical writing*. Routledge, 2006.

